

KOLO, Poland

WMO: 123450

Lat: 52.1998N

Lon: 18.6602E

Elev: 117

StdP: 99.93

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-10.5	-16.5	0.9	-12.9	-13.4	1.2	-9.9	12.3	3.2	11.3	1.4	3.3	60	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.4	22.8	30.0	21.8	28.4	20.6	24.5	29.1	23.6	27.6	22.4	26.1	4.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.1	26.9	22.3	17.2	26.4	21.1	16.0	25.5	74.4	29.4	71.1	27.6	66.5	25.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.5	7.6	-15.9	34.1	4.4	1.6	-19.1	35.3	-21.7	36.3	-24.1	37.2	-27.3	38.4		
(5)			-16.2	23.3	4.3	1.3	-19.4	24.3	-21.9	25.0	-24.3	25.8	-27.4	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	0.4	1.7	4.1	10.1	14.6	17.5	19.9	19.4	15.0	10.6	5.0	1.7	(6)
(7)		DBStd	8.72	7.55	7.13	5.29	5.44	4.54	3.66	3.62	3.24	4.13	5.28	5.21	6.40	(7)
(8)		HDD10.0	1334	319	252	197	61	9	0	0	0	3	53	165	273	(8)
(9)		HDD18.3	3267	562	470	445	257	134	58	24	26	117	252	403	518	(9)
(10)		CDD10.0	1350	21	21	13	64	152	225	306	291	154	71	16	17	(10)
(11)		CDD18.3	242	6	6	3	10	18	33	71	58	18	11	4	4	(11)
(12)		CDH23.3	2224	34	33	24	89	195	348	665	494	179	105	31	26	(12)
(13)		CDH26.7	675	8	7	8	24	49	107	225	147	56	33	7	6	(13)
(14)	Wind	WSAvg	3.8	4.3	4.2	4.4	4.1	3.6	3.4	3.4	3.2	3.5	3.7	3.9	4.1	(14)
(15)	Precipitation	PrecAvg	528	29	26	36	31	56	57	83	57	47	35	38	33	(15)
(16)		PrecMax	717	74	52	77	97	156	151	195	191	103	102	133	93	(16)
(17)		PrecMin	344	3	4	14	5	7	12	33	15	8	3	1	6	(17)
(18)		PrecStd	89	17	12	16	21	30	34	40	36	28	22	23	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	28.0	26.4	30.4	30.8	32.9	33.2	32.5	31.1	30.7	28.1	27.1	(19)
(20)			MCWB	20.9	20.1	19.4	21.2	22.3	22.1	22.5	23.1	24.2	24.6	22.9	21.6	(20)
(21)		2%	DB	23.7	24.0	19.6	26.5	28.3	30.1	31.2	30.4	28.5	27.5	22.5	22.5	(21)
(22)			MCWB	19.3	19.2	16.3	19.3	20.1	21.7	22.4	22.0	22.0	23.4	20.4	19.3	(22)
(23)		5%	DB	19.9	20.0	14.3	22.8	25.7	27.4	29.4	28.3	25.2	23.8	13.9	16.3	(23)
(24)			MCWB	17.6	17.7	9.7	17.6	18.7	19.5	20.9	20.1	19.9	21.7	11.7	14.2	(24)
(25)		10%	DB	9.9	11.8	10.9	19.5	23.1	25.1	27.2	26.2	22.8	19.2	10.9	8.7	(25)
(26)			MCWB	7.8	9.3	7.7	14.2	17.7	18.4	19.7	19.1	18.2	15.5	9.4	7.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	21.2	20.0	22.2	23.5	24.5	25.0	25.0	24.9	24.8	22.8	22.0	(27)
(28)			MCD B	26.0	25.1	25.5	27.1	28.5	29.9	30.4	30.5	29.5	29.2	26.9	25.8	(28)
(29)		2%	WB	19.9	19.8	16.2	20.6	22.1	23.0	24.1	23.9	23.8	23.6	19.8	19.6	(29)
(30)			MCD B	23.2	23.1	20.0	25.3	26.5	27.7	28.6	27.8	27.2	27.1	22.4	21.6	(30)
(31)		5%	WB	17.0	17.5	9.9	18.2	20.1	20.8	22.8	22.1	21.8	21.8	12.2	13.7	(31)
(32)			MCD B	19.7	21.0	13.3	22.8	24.2	25.9	26.9	26.0	23.4	24.1	13.6	16.2	(32)
(33)		10%	WB	8.0	9.2	7.9	13.7	17.5	19.0	20.7	20.0	17.2	15.3	9.5	7.5	(33)
(34)			MCD B	9.3	10.9	10.3	19.0	23.0	24.3	26.3	25.2	21.7	18.2	10.7	8.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.2	5.9	7.3	9.4	10.4	10.4	10.1	10.2	9.1	7.6	5.1	4.6	(35)
(36)			MCDBR	14.0	11.4	12.5	13.1	13.9	14.7	13.1	13.5	13.5	13.0	10.2	12.1	(36)
(37)			MCWBR	8.4	5.1	7.2	6.7	7.1	7.0	5.5	6.1	6.8	7.3	6.5	6.4	(37)
(38)		5% WB	MCD BR	12.9	10.6	11.1	12.3	13.4	13.9	12.3	12.9	13.0	11.9	9.7	11.8	(38)
(39)			MCWBR	7.5	4.8	7.0	6.6	7.8	7.7	6.2	6.8	8.4	7.4	6.8	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.342	0.377	0.425	0.419	0.412	0.444	0.431	0.393	0.368	0.347	0.312	(40)	
(41)		taud	2.271	2.271	2.225	2.146	2.219	2.268	2.196	2.234	2.312	2.355	2.329	2.304	(41)	
(42)		Ebn at Noon	667	759	803	806	833	842	805	793	781	721	623	604	(42)	
(43)		Edh at Noon	66	88	111	136	133	128	135	124	102	81	63	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.68	1.36	2.61	4.15	5.19	5.42	5.03	4.46	3.19	1.77	0.78	0.51	(44)	
(45)		RadStd	0.12	0.23	0.33	0.54	0.54	0.51	0.59	0.37	0.45	0.27	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-214	+106	+39
(47) Regional (21 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-170	+82	+28

Nomenclature: See separate page